

Short communication

Failure behaviours of 100% SOC lithium-ion battery modules under different impact loading conditions

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ABSTRACT

To support the development of failure prediction models and improve the crash safety designs of the energy storage system of electric vehicles, a type of battery module consisting of pouch cells for commercial use is tested under impact loading, which is an extreme loading type of mechanical abuse. Using a test matrix designed for modules with 100% SOC, a series of impact tests are performed on a drop tower with various combinations of loading speed, drop mass, orientation of the module, and shape of the impactor. The mechanical, electrical, and thermal responses of the modules under various loading conditions are analysed. By comparison between cases, the influence of the loading conditions on the deformation and damage modes and the likelihood of short circuit occurrence and thermal runaway are discussed.

1. Introduction

Lithium-ion batteries are widely used as energy storage devices for electric vehicles, taking advantage of their high energy density and recharging ability. The battery cells are usually grouped into small modules, which are then packed together with cooling and management systems to form the battery pack of electric vehicles. The fire accidents induced by vehicle crash that occasionally occurred in recent years have drawn significant attention on the safety of lithium-ion batteries under impact loading.

To clarify the underlying mechanism and improve the numerical prediction capability for lithium-ion battery failure under crushing or impact loading, previous researches focus their investigation on the different structural levels of the battery. On the battery cell level, Wierzbicki and co-workers [1,2] and Greve and Fehrenbach [3] conducted quasi-static tests on both cylindrical and pouch cells under different loading and boundary conditions and revealed the relationship between the onset of internal short circuit and the features of mechanical response. Wierzbicki and co-workers [4,5] built an analytical model for homogenized characterization of jellyrolls and finite element models of battery cells. Wang et al. [6] put forward a simple analytical clay-like model to characterize the radial direction compression of cylindrical battery cells. Greve and Fehrenbach [3] incorporated the Mohr-Coulomb fracture criterion into the battery cell model and validated the simulation of various mechanical loading cases in each temperature profile of battery cells monitored by an infrared camera. Xu et al. [7–10] investigated the mechanical integrity of 18,650 cylindrical battery cells under different loading conditions (e.g. nail penetration, compression and bending), and established a rate-dependent and SOC-dependent homogenized model for cylindrical cells. Pan and co-workers [11] developed a representative volume element model of pouch type batteries, which was then employed to analyse the development of local deformation patterns inside the battery under in-plane compression. Meanwhile, other researchers, such as Maleki et al. [12,13] and Lamb and Orendorff [14] conducted tests of battery cells under specific loading conditions, such as pinch, penetration and bending, to acquire mechanical responses and analyse

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the influence of different factors on the internal short circuit of battery cells.

Currently, most of the studies on mechanical response are focused on the battery cell level. The work on the battery module level or above is mainly concerned with thermal behaviour or electrical management. Lamb et al. [15] monitored the thermal runaway propagation of small battery modules with different cell types and electrical configurations after a hard internal short circuit induced by mechanical nail penetration in one of the cells. Feng et al. [16] also triggered the short circuit using nail penetration and tested the thermal runaway in a large-form lithium-ion battery module consisting of six prismatic batteries. They found that the heat transferred through the battery shell dominates the thermal runaway propagation process. Zhao et al. [17] proposed another electro-chemical–thermal model according to nail penetration tests, and in their study, a hydrogel cooling system proves the effectiveness in preventing battery thermal runaway. A pilot study on the safety and protection of the battery pack in electric vehicles against ground debris impact was conducted by Xia et al. [18]. In this study, the local indentation into the battery pack consisting of cylindrical cells was simulated, and the deformation and damage evolution of individual battery cells and influence of the fracture in shield plates made of different metal materials were analysed.

Very few publications are found to report the mechanical behaviour as well as the associated electrical-thermal response of lithium-ion battery modules under impact-type mechanical loading. To validate the numerical model built for predicting battery failures at various levels, test results at the module level are necessary. Moreover, comprehensively understanding the mechanical responses, the possibility of short circuit onset and thermal runaway at the module level under various impact loading types is fairly helpful for the improvement of the battery pack architecture and the protective countermeasures. With these purposes, the present study is focused on a type of battery module consisting of pouch cells for commercial use. It designs and conducts a series of impact loading tests for the modules with 100% SOC, distinguishes the main difference of the mechanical, electrical and thermal behaviours among the modules under various loading conditions, and elucidates its underlying mechanisms or effects.

2. Experiments

2.1. Description of the tested battery module

The pouch cell in the tested battery module is a type of battery containing $\text{LiNi}_x\text{Mn}_y\text{Co}_z\text{O}_2$ cathode, which achieves 3.7 V nominal voltage and 20 Ah capacity at 100% SOC. The charging-discharging curves of four pouch cells are shown in Fig. 1, where six charging-discharging cycles are applied to each cell. The variation of charging or discharging time is less than 2.6%.

The dimensions of a single battery module in the present study are shown in Fig. 2a. The composition of the battery module is detailed in Fig. 2b. The module mainly consists of four lithium-ion battery units. In each battery unit, two pouch cells in parallel connection are fully stacked on each other. An aluminium cooling fin is sandwiched in between, and four pieces of plastic frames are assembled by snap-fit to fix the pouch cells and the cooling fin. Each battery unit has a positive and a negative electrode. The four battery units are packed together with two plastic covers on the top and bottom side by bolt fastening.

The module is configured with a 4P2S layout, as shown in Fig. 2c. Therefore, the nominal voltage of the module is 7.4 V and the capacity is 80 Ah at 100% SOC. All modules were fully charged prior to testing.

The orientation of battery modules in an electric car is illustrated with the floor architecture of the battery pack in Fig. 2d. For individual modules, their X or Y direction is associated with side impact or frontal crash scenarios, while their Z direction is associated with bottom impact scenarios.

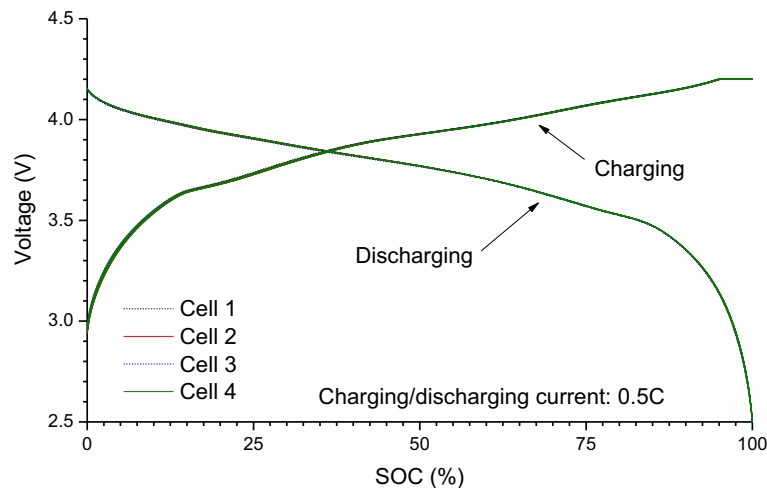


Fig. 1. Charging-discharging curves of pouch cells.

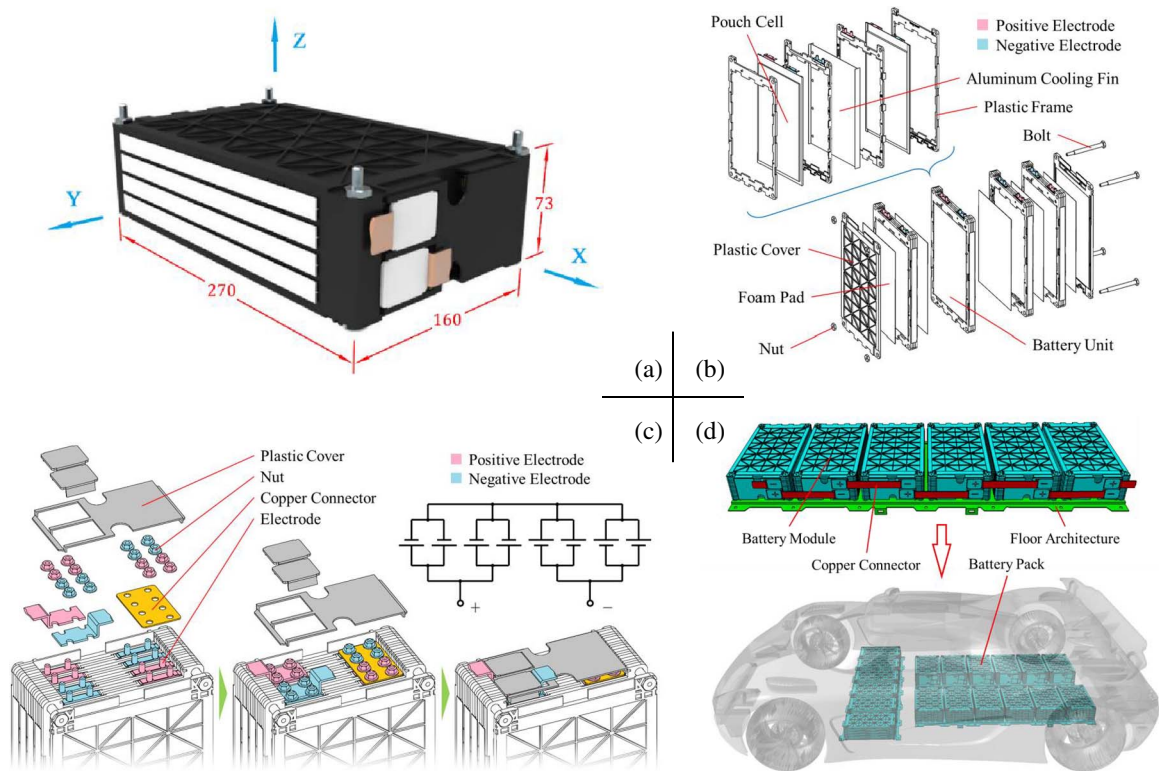


Fig. 2. Detailed structure of the battery module: (a) dimensions, (b) explosion view of a unit in the module and assembly of the units, (c) electrical connection in the module, and (d) illustrated layout of the battery modules in electric vehicles.

2.2. Setup and loading condition of module impact tests

All impact tests in the present study were conducted with a drop tower built in-house, which has a net height of 7 m and an adjustable drop mass ranging from 20 to 500 kg, as shown in Fig. 3a.

The battery module was mounted in the testing room just below the drop mass. Fig. 3b shows the method of fixing the modules to be impacted along the X, Y and Z directions. The four corner bolts were used to fix the battery module during the impact tests. In the X-direction or Y-direction impact tests, the module stood along its X-axis or Y-axis above a support plate (a thick steel plate shown in blue in Fig. 3b) with each bolt at the lower end being firmly clamped by two steel hooks. In the Z-direction impact tests, the module was directly fastened by the four bolts on the support plate.

Between the support plate and the rigid foundation, a customized force sensor made of steel was mounted, as shown in Fig. 3b. Strain gauges were glued on the side walls of the sensor, with which Wheatstone bridges were configured to detect the voltage change due to external mechanical loads. A four-channel dynamic strain amplifier was employed to record the voltage signals from the strain gauge based force sensor.

Given that the voltage of the battery module itself does not need amplification, it was directly monitored during the impact test via high-speed AD data acquisition card. The sampling rate of all voltage signals is 1 MHz.

The intrusion distance of the impactor during the impact test was measured using noncontact method. Random speckles were beforehand made on the side surface of the impactor. A high-speed digital camera captured a sequence of images during the intrusion process with a frame rate of 10,000 fps. The displacement of the impactor was then calculated by tracking the motion of speckles via digital image correlation (DIC). The image sequence also provided information of the local deformation and fracture in the impacted battery module.

A test matrix was designed and executed in the present study, as shown in Table 1. The test matrix is far from a full factorial design of experiments due to the limited number of test samples and the costly preparation of drop-tower impact tests. However, it is still possible to evaluate the influence of different loading conditions on the impact response of battery module with the present results.

2.3. Mechanical properties of major components in the battery module

The load in the battery module test is mainly undertaken by the battery cells, the plastic structures and the aluminium cooling fins. Quasi-static and dynamic mechanical tests are performed with specimens cut from these major components to identify their

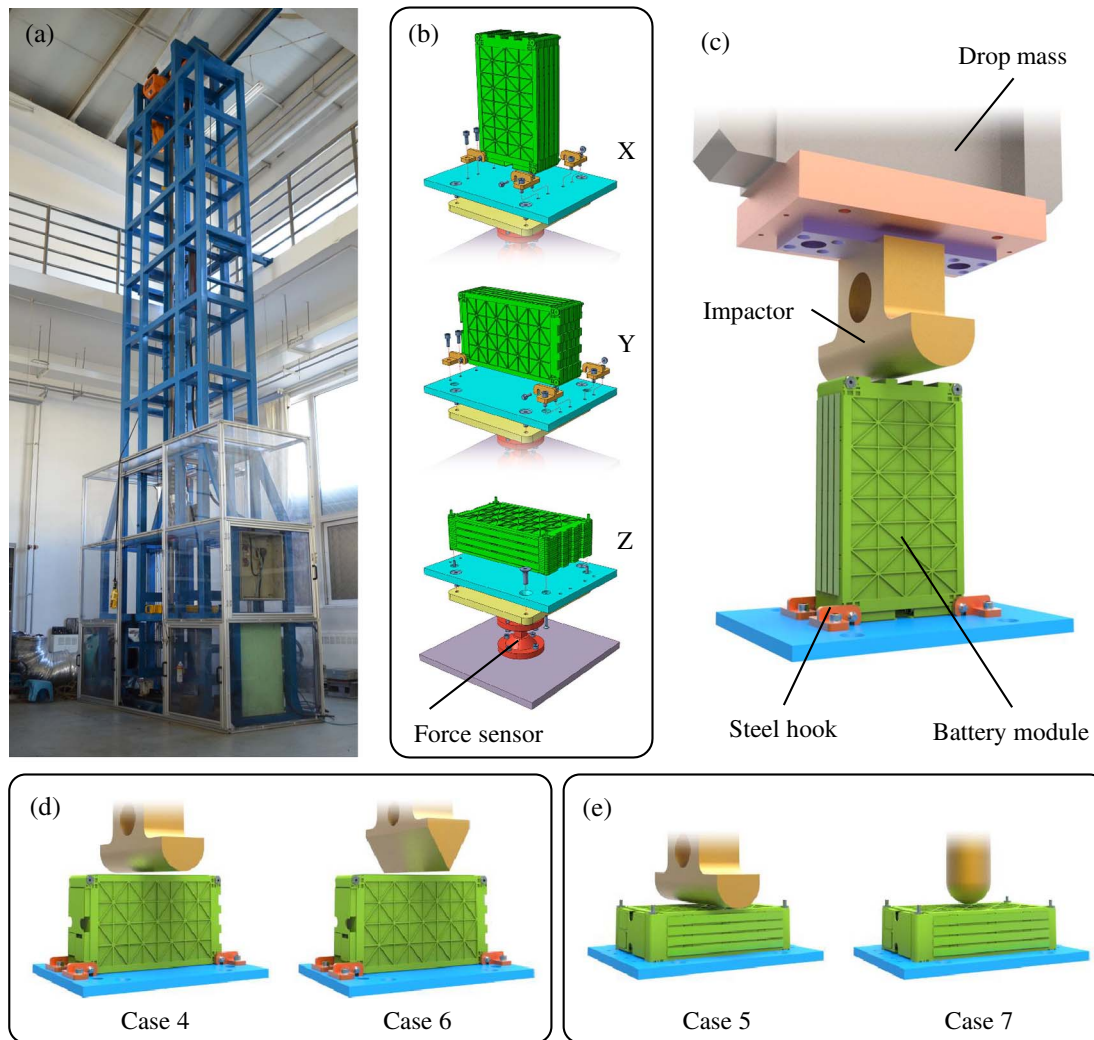


Fig. 3. Test setup of the module impact tests: (a) drop tower, (b) fixtures for impact along different directions, (c) X-direction impacts, (d) Y-direction impacts with different impactors, and (e) Z-direction impacts with different impactors. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

Table 1
List of battery module impact tests.

No.	Shape of impactor	Impact direction	Mass (kg)	Velocity (m s^{-1})
1	Cylindrical	X (length)	81.6	10
2	Cylindrical	X (length)	81.6	7
3	Cylindrical	X (length)	40.0	10
4	Cylindrical	Y (width)	40.0	10
5	Cylindrical	Z (thickness)	40.0	10
6	Wedged	Y (width)	40.0	10
7	Spherical	Z (thickness)	40.0	10

basic material properties, which are beneficial for understanding the different mechanical responses of the battery module in various loading cases.

The plastic material of the battery module covers and frames presents the isotropic strain-rate dependent elastoplasticity (Fig. 4a), and the comparison among the tension, compression and shear equivalent stress-strain curves indicates the hydrostatic pressure dependence of the elastoplasticity (Fig. 4b). The aluminium material of the cooling fins apparently presents isotropic and approximately ideal elastoplasticity that meets the von Mises yielding criterion (Fig. 4a).

As the smallest unit of function and integrity in the module, the battery cell is tested separately. To compare the difference

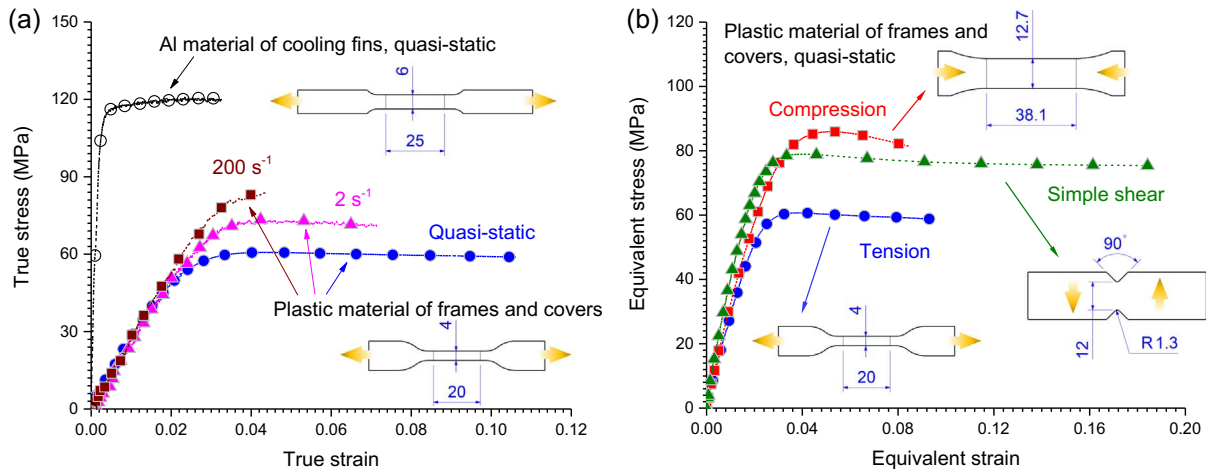


Fig. 4. Mechanical test results of materials: (a) uniaxial tension of plastic structure material and aluminium cooling fin material, and (b) tests at different stress states of plastic structure material.

between in-plane and out-of-plane mechanical properties, square specimens with dimensions of 25×25 mm are cut from the pouch cells and subjected to compression tests. As shown in Fig. 5a, the load-bearing capacity along the out-of-plane direction is much stronger than that along the in-plane direction. The indentation tests of the single battery cell are also performed, as shown in Fig. 5b, where the loading rate effect of the out-of-plane properties can be apparently observed. Before the mechanical failure, the stiffness (initial slope of the force-intrusion curve) under the dynamic load is larger than the counterpart of the quasi-static case. An interesting phenomenon is that the first force peak (corresponding to the onset of the mechanical failure inside the pouch cell) under dynamic indentation is quite earlier and smaller than that under quasi-static, probably due to the more localized deformation concentration and less ductility of the critical constituents (especially, the separator) under the dynamic load.

3. Results of battery module impact tests

The test of each case in Table 1 was only repeated twice because of the limited number of test samples. Fig. 6 shows the repeated test results of Case 2, where the repeatability is acceptable regarding the complicated structure of the battery module.

At the final stage of the impact, the dropping mass contacted the bumpers, which were used to protect the force sensor, and the remaining kinetic energy of the dropping mass was absorbed by the continuous densification, fragmentation in the tested battery module and the accumulated deformation of the bumpers. Therefore, the oscillation of reaction force detected by the force sensor becomes more severe and irregular, as shown in grey curves in Fig. 6.

For a clear comparison in the remaining part of the present paper, a five-point smoothing method was applied to each force-time curve or force-displacement curve as shown in red and orange curves in Fig. 6, where it can be observed that the smoothed result almost coincides with the raw curve except that the huge oscillation at the final stage is eliminated. Furthermore, in the remaining

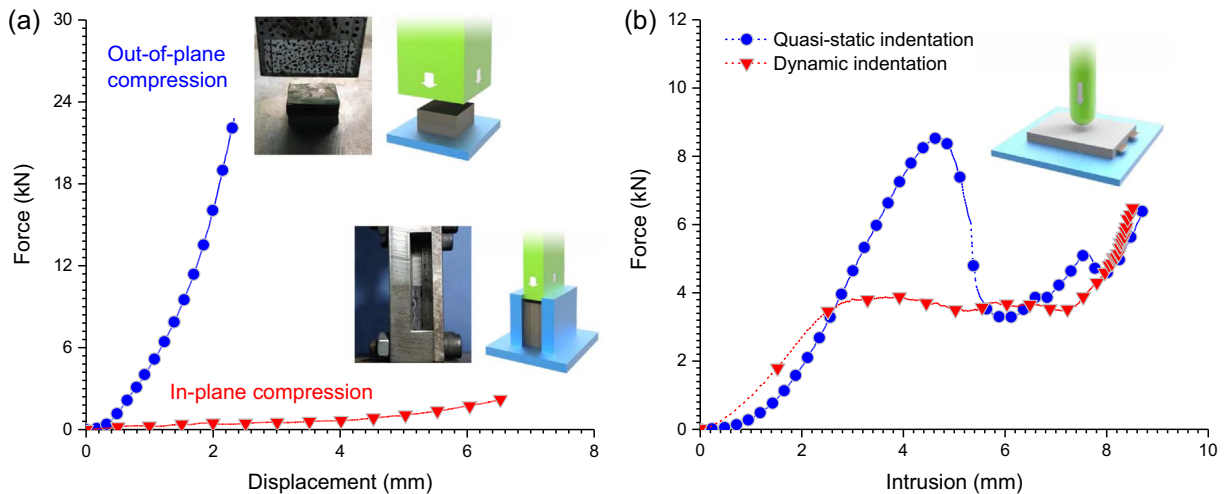


Fig. 5. Mechanical test results of the battery cell: (a) anisotropic behaviour, and (b) indentation at different loading rates.

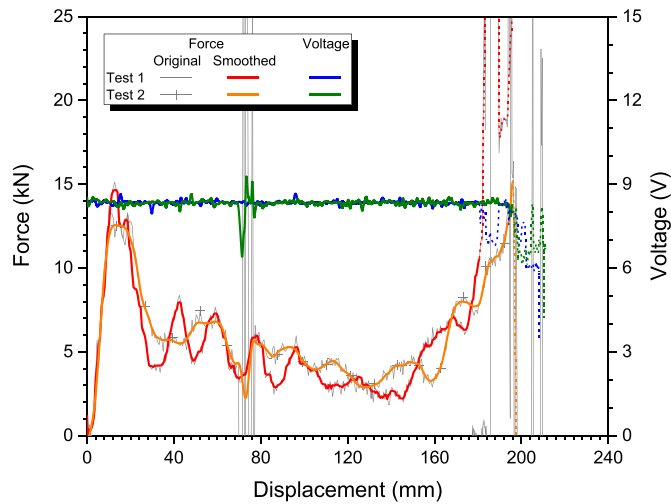


Fig. 6. Repeated test results (Case 2).

part of the present paper, only one representative result is plotted for each case, and the stage after the bumper contact is eliminated in the curve plot.

3.1. Influence of mass and initial velocity (Cases 1–3)

The force, displacement and voltage measured over time in Cases 1–3 are shown in Fig. 7a, while the force and the mechanical energy consumed over the intrusion distance in the three cases are shown in Fig. 7b. The three cases present similar characteristics

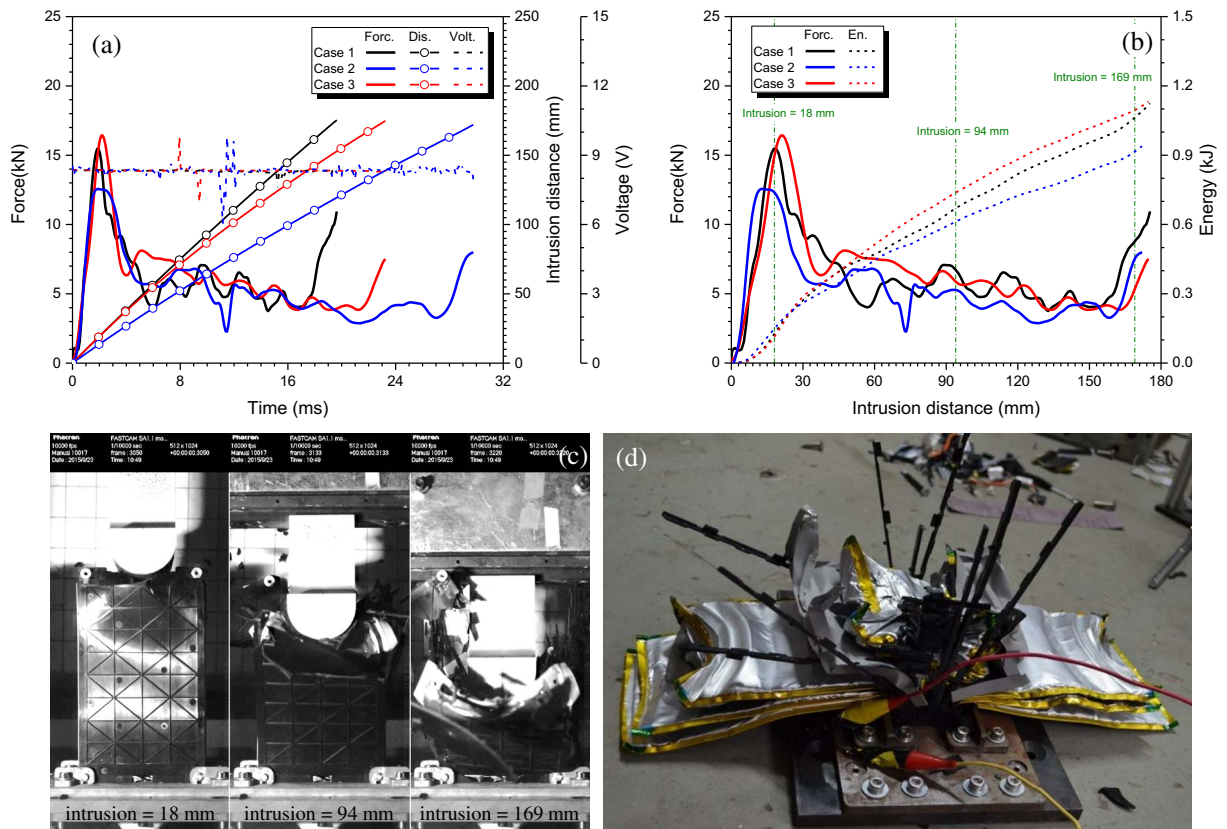


Fig. 7. Test results of Cases 1–3: (a) mechanical and electrical responses over time, (b) mechanical responses over intrusion distance, (c) process of battery module deformation and fracture in Case 1, and (d) post-mortem battery module.

regarding the responses in Fig. 7a and b. At the beginning of the X-direction impact, the assembly of plastic frames, covers and the aluminium cooling fins, referred to as the frame structure, mainly bears the impact loading, while the battery cells play a secondary role due to their lower stiffness along the X-direction (as demonstrated in Fig. 5a). At the initial stage (intrusion distance about 0–40 mm), as exhibited with the test results, the highest peak of the impact force is generated to trigger the initial buckling of the frame structure. Within the intrusion distance of about 40–160 mm, a series of force fluctuations are generated at a lower level due to subsequent buckling and fracturing in the frame structure, and the gradual involvement of the battery cells in load-bearing with the form of bending and folding. From the intrusion distance of about 160 mm, the force response starts to rise significantly due to the compaction of individual battery cells under the impactor head.

The high-speed camera also recorded the deformation and fracture process during the battery module impact test. Taking Case 1 as an example, Fig. 7c displays the deformation and fracture pattern of the battery module at the typical moment in each of the three stages.

At the very late stage of the impact, owing to the severe breakage of the frame structure, the battery cells lose the restraint and fell towards two sides, as shown in Fig. 7d.

As shown in Fig. 7a, the impact with the largest initial kinetic energy (Case 1) in the three cases has the shortest time duration to achieve the densification. For Cases 2 and 3, having equal initial kinetic energy, the impact with the larger initial velocity (Case 3) achieves the densification earlier.

As shown in Fig. 7b, the first force peaks of Cases 1 and 3 are relatively close to each other, which are mainly relevant to the same initial impact velocity in the two cases. Comparatively, the first force peak of Case 2 is apparently lower because of the lower initial impact velocity.

In all of the three cases, the voltage keeps constant despite the module undertaking a large intrusion up to 65% of the X dimension, as shown in Fig. 7a. The voltage did not drop until the very late stage of the impact. As shown in Fig. 6, a short circuit occurs at the stage after bumper contact when individual battery cells are broken or the electrode terminals connect with each other under the extreme squashing. However, in all of the three cases, no thermal runaway of the module is observed because the final collapsing separation provides the short-circuited module a good condition of heat exchange with the environment.

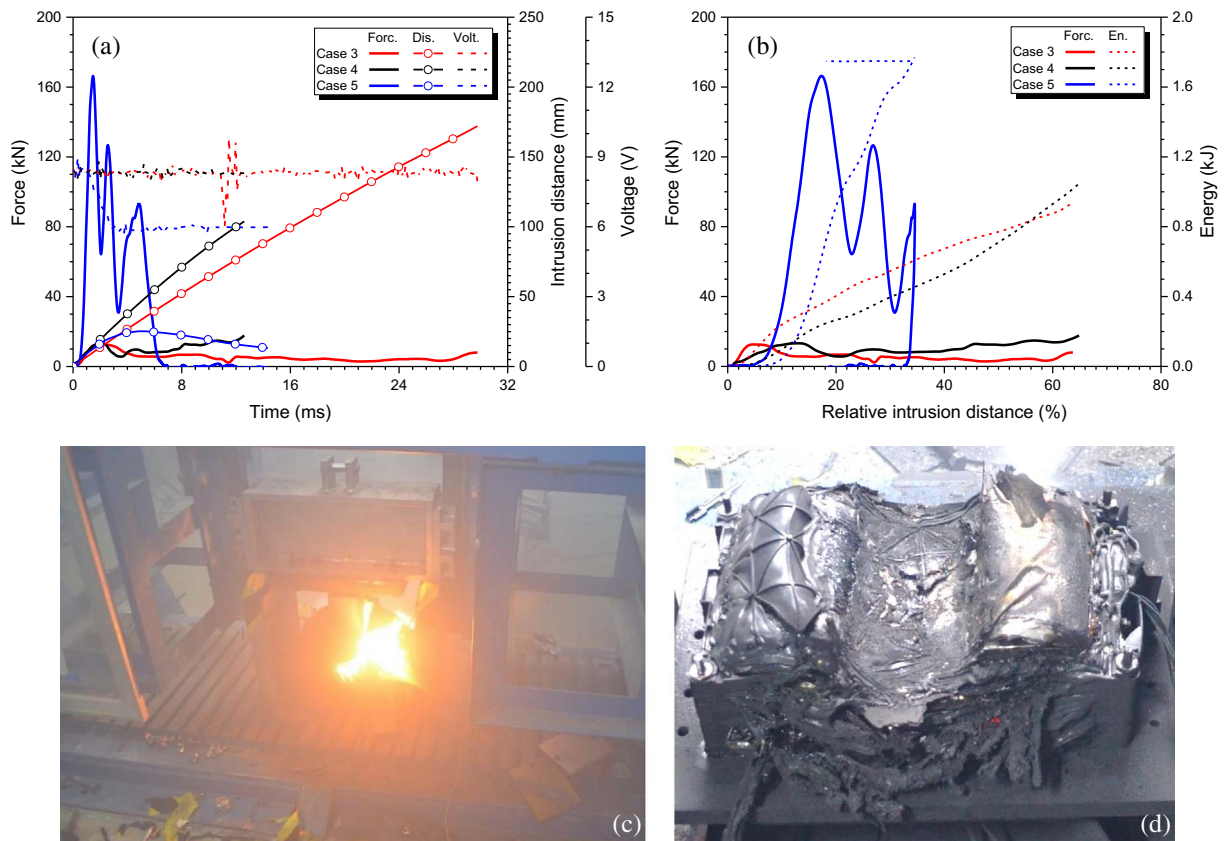


Fig. 8. Test results of Cases 3–5: (a) mechanical and electrical responses over time, (b) mechanical responses over relative intrusion distance, (c) smoke and fire in the Z-direction test, and (d) post-mortem sample of the Z-direction test.

3.2. Responses of battery modules impacted from different directions (Cases 3–5)

Fig. 8a shows the force vs. time responses, the intrusion distances over time and the voltages over time of the battery modules impacted from the X-, Y- and Z-direction, respectively. Fig. 8b shows the force and the mechanical energy consumed over the relative intrusion distance in the three cases. Given that the X-, Y- and Z-dimension of the battery module are different, the mechanical response is better compared along the three directions based on the relative intrusion distance instead of the absolute distance. The relative intrusion distance is a ratio of the absolute intrusion to the full length in the test direction.

The force-time plots show that the impact force level in the Z-direction is significantly larger than that of the X- or Y-direction, which implies a substantial difference of the impact resistance mechanism between the Z-direction and the other two directions, essentially reflecting the structural characteristics of the battery module.

For the impacts along the X- and Y-direction, the first force peaks are close to each other, while the average plateau force level after the first peak in the Y-direction impact is slightly higher than that in the X-direction impact. The difference is mainly caused by the extent of the separation between cells in the module during the plateau stage.

The initial stiffness along the Y direction is also close to that along the X direction. Note that in Fig. 8b, the initial slope of Case 3 (X-direction impact) looks apparently larger than that of Case 4 (Y-direction impact) because the abscissa is the relative intrusion distance and the X-direction span of the module is larger.

The intrusion distance progression shows that compared to the other two cases, the impactor rebounds much earlier in Case 5 (Z-direction impact) without contacting the bumpers. Specifically, the Z-direction of the battery module provides a much larger resistance to intrusion, and the deformation, cracking and compacting during the intrusion of the impactor consume all of the initial kinetic energy at the maximum intrusion distance. For the other two cases, the resistance to intrusion is much lower, and consumption of the impact energy is very limited before the impactor contacts the bumper.

The voltage progression shows that the impact along the Z-direction is the most severe case in terms of short circuit and thermal runaway. An apparent drop of the voltage is observed only in the Z-direction impact. Although the relative intrusion distance in this direction was only about half of the other two cases, a severe thermal runaway was triggered in the test. The battery module started to smoke rapidly after the impact and eventually resulted in a fire, as shown in Fig. 8. The battery module expansion and the gas venting can even lift up the 40 kg impactor because of the thermal runaway. Fire and smoke were generated in all tests of the Z-direction impact in the present paper.

Although no object directly penetrated into the battery module in Case 5, the stacked battery cells under the out-of-plane impact with a large kinetic energy suffered internal crack initiation and propagation severely. The jellyroll structure in the cells was crushed, and the anodes and cathodes exposed to one another locally when the in-between separators were ruptured. The rapidly increasing current caused by an internal short circuit generated heat at the local areas intensively, which cannot be exchanged with the environment in time. The temperature eventually increased and then ignited the flammable constituents in the battery cells.

3.3. Responses of battery modules loaded by impactors with different shapes

3.3.1. Y-direction impact with sharp and blunt impactors (Cases 4 and 6)

The force-time or force-intrusion curves of the two tests have similar profiles, as shown in Fig. 9a and b, where the case with sharp impactor (Case 6) overall presents a lower force level and a smaller slope of the initial force compared to the case with blunt impactor (Case 4). However, the failure modes are entirely different between the two cases, as shown in the crushing processes recorded by the high-speed camera in Fig. 9c and d, respectively.

With the post-mortem samples, the difference of the failure between the two cases can be more clearly observed in terms of the deformation mode and the fracture pattern occurring in the multi-layer structure of the module, as shown in Fig. 9e and f. In Case 6, the cells and the cooling fins were cut through by the sharp impactor, together with bending as the major deformation mode and a little amount of buckling near the cut location. In contrast, the multi-layer structure impacted under the blunt impactor in Case 4 was folded into large wrinkles without apparently observable cracks. The failure mode of Case 4 can absorb more mechanical energy, as shown in Fig. 9b, because more materials of the module were engaged into large deformation.

The voltage developments in the tests were also recorded, as shown in Fig. 9a. In the case with the blunt impactor (Case 4), the voltage does not drop until the densification stage, while the case with the sharp impactor (Case 6) shows a very early drop and a continuous decrease of the voltage. As described previously, the anodes and cathodes in the broken cells cut by the sharp impactor are easily exposed to one another, leading to the early occurrence of the internal short circuit. Meanwhile, the components in the module tend to separate from one another under the impact along the Y-direction; therefore, the connection of electrodes remains unstable. Along with the successive contact and separation in the multi-layer structure, the voltage of the module decreases gradually, not abruptly. Owing to the relatively long process of energy release and the significant tendency of the components to separate, no thermal runaway occurred during the tests of Case 6.

3.3.2. Z-direction impacts with cylindrical and spherical impactor (Cases 5 and 7)

The impact tests along the Z-direction with cylindrical and spherical impactor show similar characteristics in the force-time or force-intrusion responses, as shown in Fig. 10a and b. After a short term of small slope corresponding to the initial impactor-module contact that eliminated the initial gaps inside the multi-layer structure, the slope increases dramatically, exhibiting a large resistance of the module against the impact along the Z-direction. After the first force peak in the response profiles, along with the occurrence of the unstable deformation or local fracture in the upper components of the multi-layer structure and the successive engagement of the

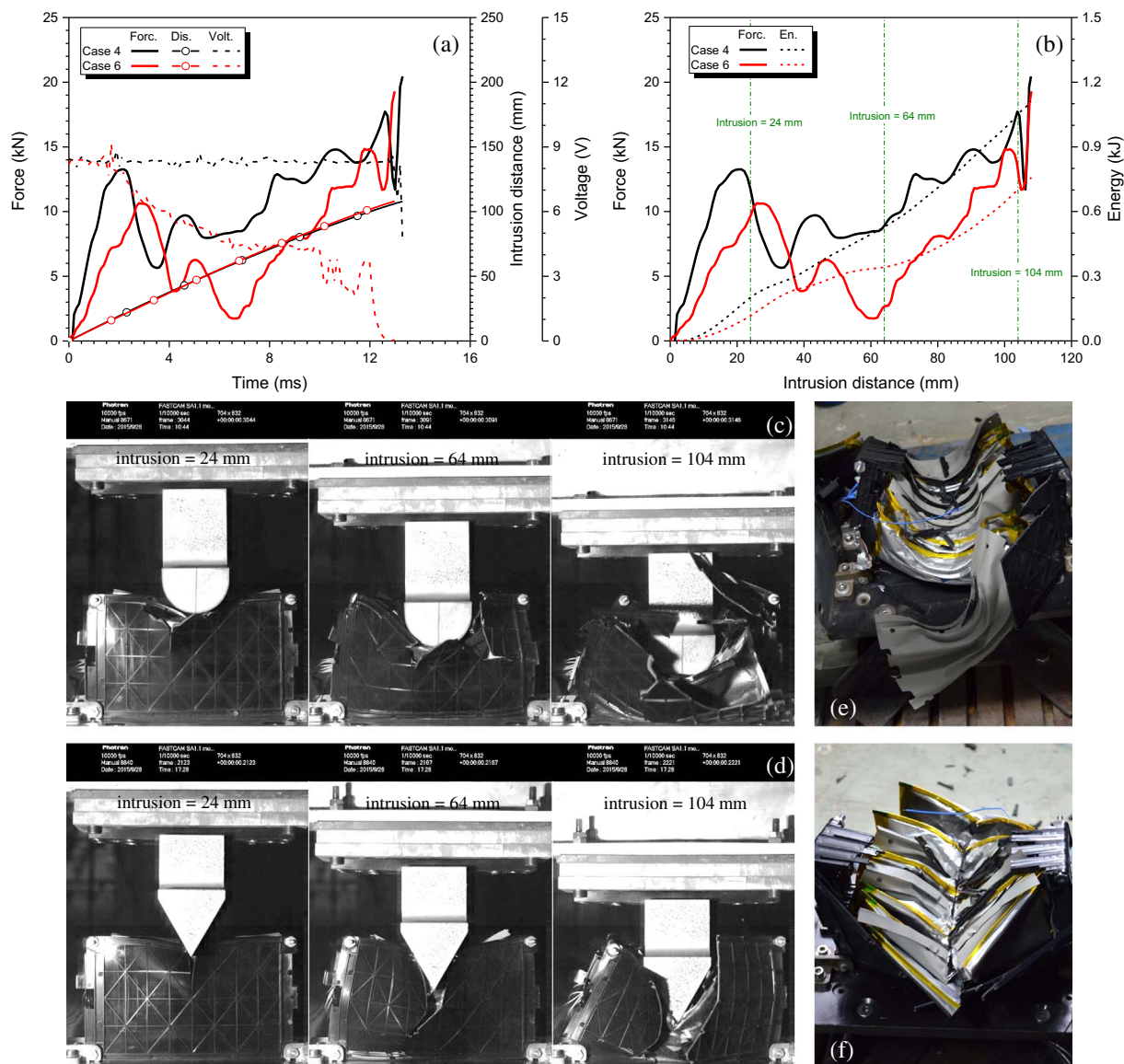


Fig. 9. Test results of Cases 4 and 6: (a) mechanical and electrical responses over time, (b) mechanical responses over intrusion distance, (c, d) crushing processes of the battery modules in Cases 4 and 6, and (e, f) post-mortem samples of impacts with a blunt cylindrical impactor and a sharp wedged impactor.

lower components in load-bearing, force crests and valleys were formed. Internal short circuit occurred eventually after the impact started in both cases, as shown in the voltage developments in Fig. 10a. Until the initial kinetic energy of the impactor is totally consumed, the maximum intrusion is achieved, and the impactor is rebounded. Fig. 10c and d shows the photographs recorded by the high-speed camera at the moment of force crests in Fig. 10b (marked with hollow dots), where the process of destroying the module by the impactor can be observed in sequence.

Compared to the intrusion with cylindrical impactor (Case 5), intruding the module along the Z-direction with a spherical impactor is much easier; therefore, the impact force level in Case 7 is smaller and the total intrusion distance is larger, as shown in Fig. 10b.

With the battery cells being fractured and compacted in the penetration, all of the battery modules tested in Case 7 finally suffer severe thermal runaway to ignite fire intensively. The fire and smoke scene, as well as the post-test battery module, is shown in Fig. 10e and f.

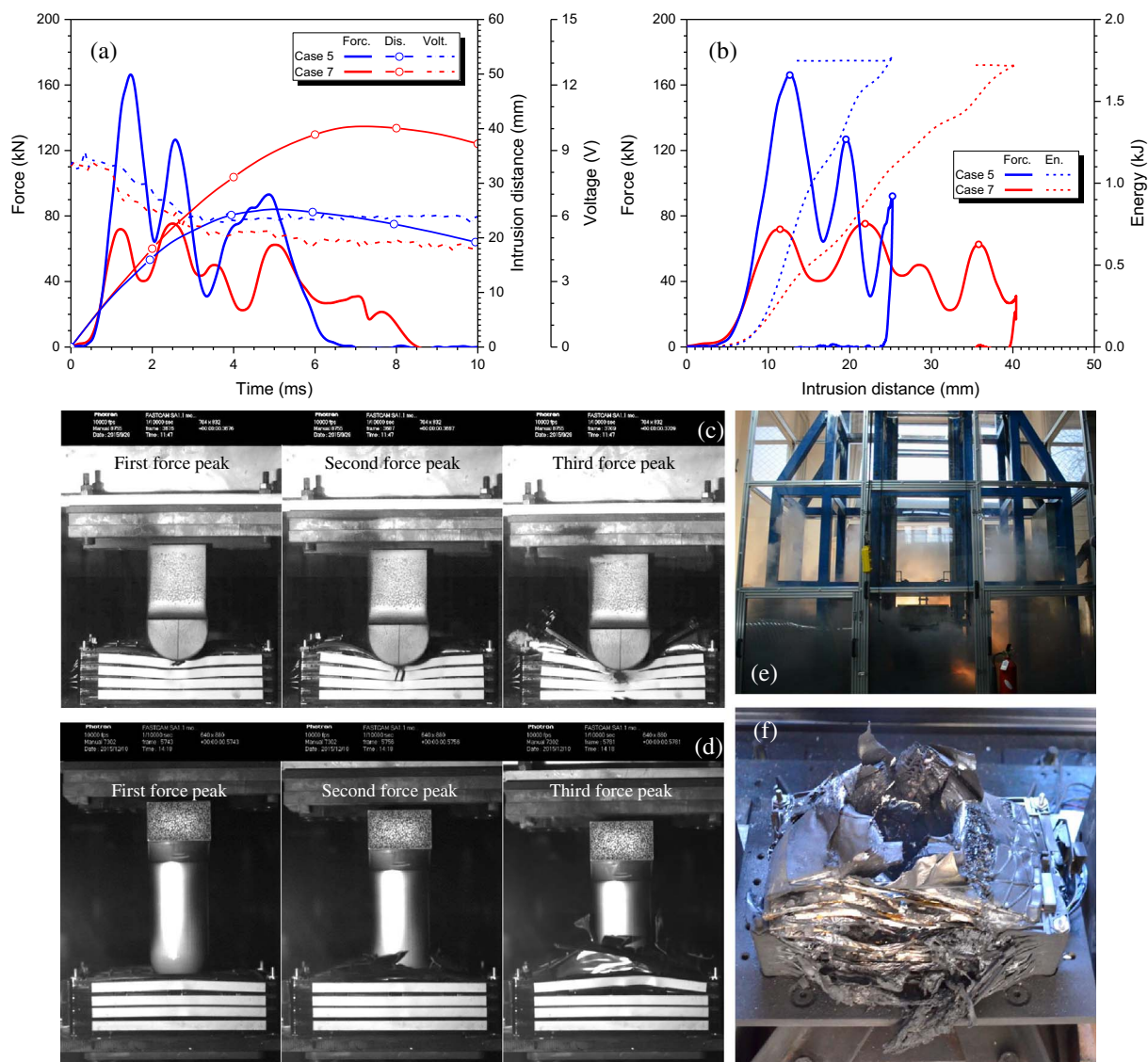


Fig. 10. Test results of Cases 5 and 7: (a) mechanical and electrical responses over time, (b) mechanical responses over intrusion distance, (c, d) processes of destroying the battery modules in Cases 5 and 7, (e) fire and smoke scene, and (f) the post-test battery module.

4. Discussion

4.1. Influence of SOC on response to impact loading

All of the test results presented in this study are focusing on the 100% SOC battery modules. The influence of SOC on the response to impact loading has not been fully investigated in the present study, although a few tests on 30% SOC battery modules were performed. The test condition was set similar to that in Case 2, except for SOC. As shown in Fig. 11, the comparison between the two cases indicates that for the fully-charged and partially-charged modules under X-direction impact with cylindrical impactor, the global mechanical responses, as well as the overall developments of voltage, are similar to each other. No thermal runaway occurred in both cases.

However, it cannot be concluded that there is no difference in terms of the thermal response for the battery modules with different SOC. The frame structure is ruptured and the inside cells are separated from one another during the impact along the X or Y direction, making heat transfer easier between the cells and the environment, which greatly diminishes the likelihood of thermal runaway. If the impact tests are performed along the Z direction, the situation of thermal response should be entirely different. The heat accumulation should be more intensive in the battery module with larger SOC when the cells are compacted tightly, leading to higher risk of thermal runaway, or more severe thermal runaway. This scenario deserves a quantitative analysis when the number of test samples is adequate.

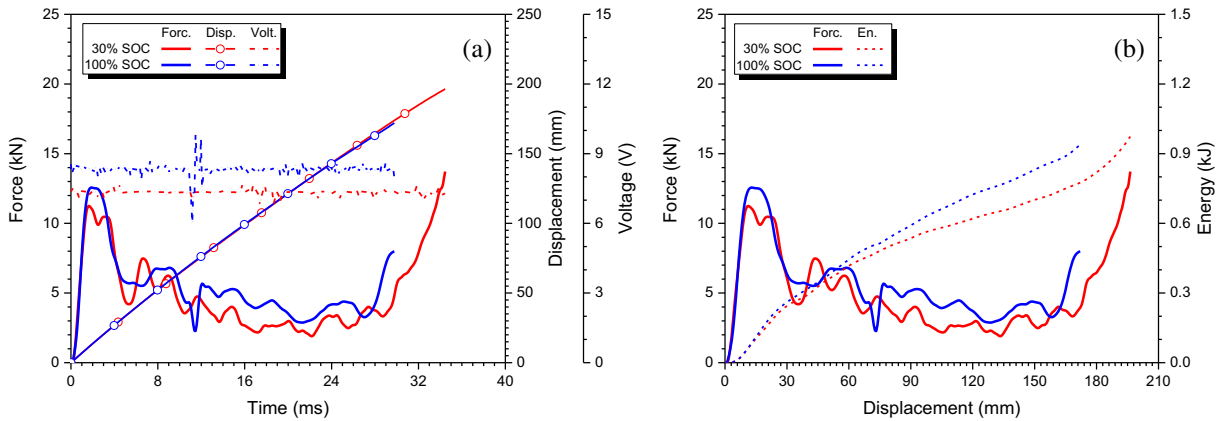


Fig. 11. Test results of battery modules with 30% SOC and 100% SOC: (a) mechanical and electrical responses over time and (b) mechanical responses over intrusion distance.

4.2. Boundary conditions and impact severities

From the present study, it can be concluded that impact intrusion along the thickness direction of the battery modules is very dangerous and tends to induce severe thermal runaway. Electric vehicles adopting the layout of the battery pack as shown in Fig. 2d are in need of careful design of the protective countermeasure to cope with the local impact from ground protrusions or debris.

Impact tests along the other two directions in the present study did not show thermal runaway even though the modules were crushed eventually in a destructive manner and short circuit also occurred. As described previously, the major reason for this scenario is the fracture mode of the module providing effective condition of heat transfer. However, considering the realistic installation of the modules in a battery pack and the surrounding components in the vehicle body, fragments and components of a battery module cannot be separated or scattered from one another so freely during the impact. Therefore, the free boundary condition at four sides of the module in the tests actually deviates from the reality. When the boundary condition is modified, it is inferred that the likelihood of thermal runaway becomes larger after the impact along the X or Y direction.

The discussion on the boundary condition of the tests along the X or Y direction also provides a little information for the safety design of the battery module/pack structure. If the battery modules have to undergo severe crushing along either of the two directions, the appropriate trigger of casing fracture and components separation can be considered because these mechanisms can take part in the impact energy dissipation and provide room for heat transfer.

5. Conclusions

Mechanical responses of a commercial lithium-ion battery module under various impact loading conditions, as well as the likelihood of thermal runaway after the impact, are evaluated in the present study. Knowledge for the impact tests of small components as well as the measurement techniques of the dynamic responses accumulated in the investigators' group are well employed in establishing the impact test platform for battery modules. The following conclusions can be drawn based on the comparative analysis of the impact test results of the 100% SOC modules.

- (1) In the impact tests along the X- or Y-direction, most of the load-carrying capacity of the battery module loses when the frame structure starts to fracture. Then, the rest of the frame structure fails continuously, and the battery cells are gradually deformed by buckling and bending. In the impact tests along the Z-direction (i.e. the cell stacking direction), the battery module can carry larger load and absorb more energy. With the frame structure fractured gradually and the battery cells compacted tightly, peaks are successively generated in the force response.
- (2) No thermal runaway occurred after the impact tests along the X- or Y-direction despite the short circuit because of the good condition of heat transfer led by the destructive failure of the frame structure and separation between the battery cells by disassembly. Comparatively, impact along the Z-direction is quite dangerous. Given the increasing compaction and tightness during the impact, the battery cells fail severely, and the heat generated is not released effectively, leading to thermal runaway in the form of fire and smoke.
- (3) To estimate the risk of thermal runaway for a battery module under typical impact load types, two aspects should be considered. The first is whether the main deformation mode of cells easily triggers internal short circuit, and the second is to what extent the interplay between cells blocks the heat exchange with the environment during and after the impact. These considerations are suitable for any type of lithium-ion battery module. It also provides clues for designing or optimizing safe configurations of battery modules against impact loads.

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